

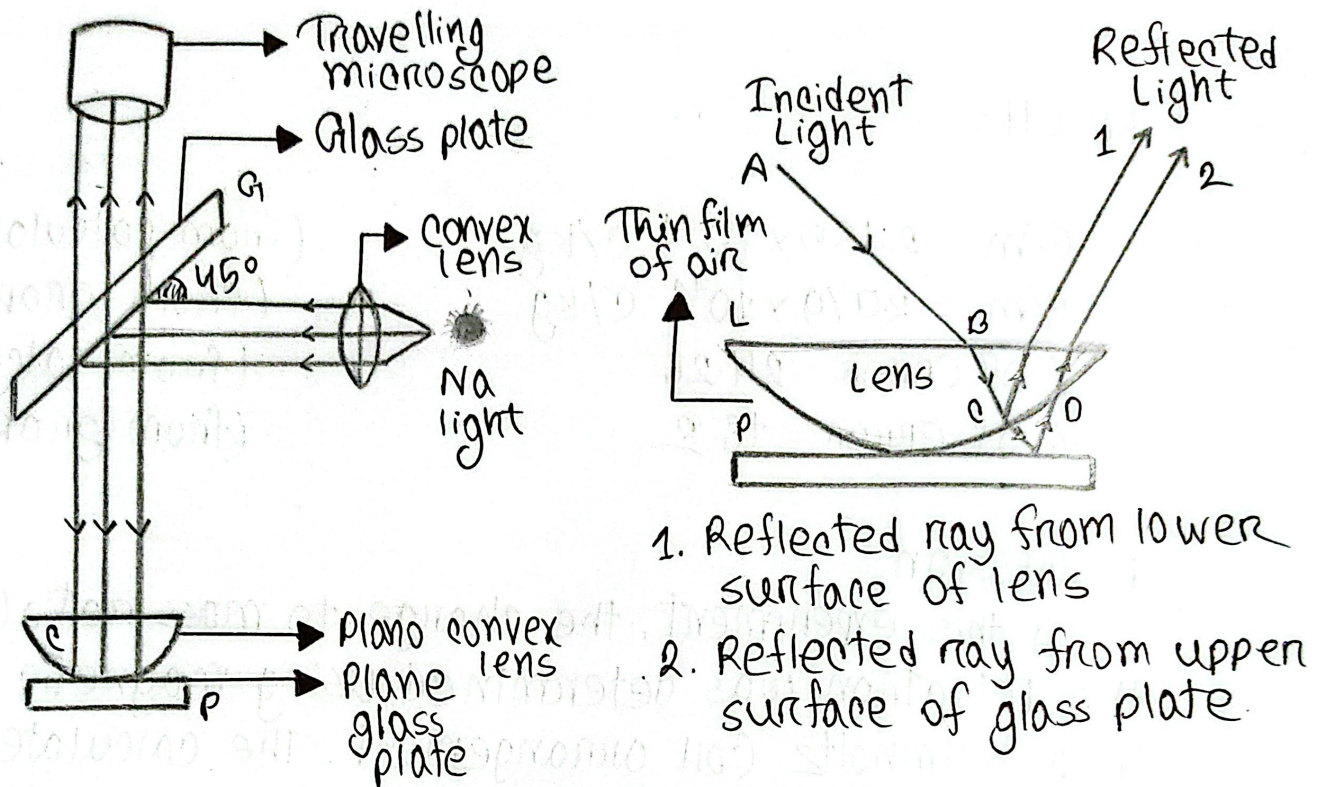


Fig 1: Experimental setup to find Newton's Rings

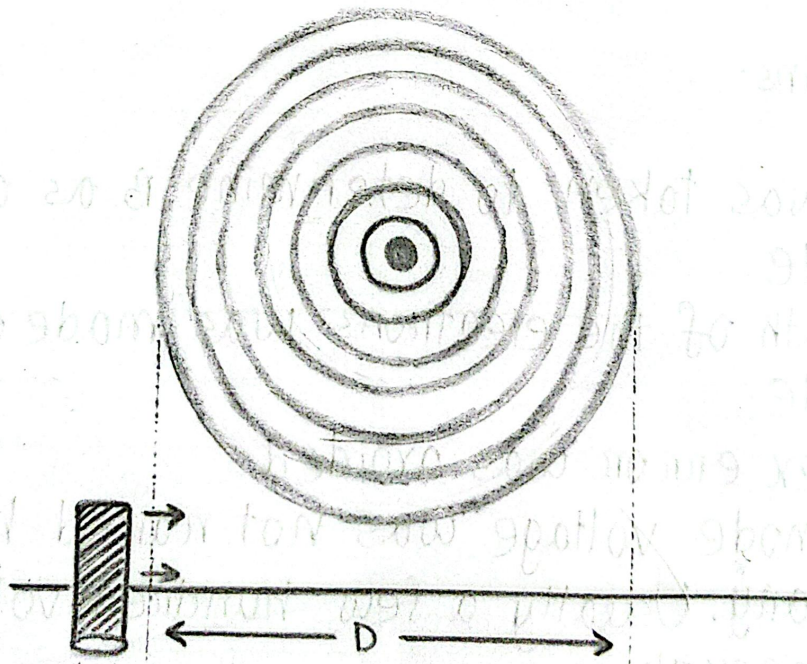


Fig 2: Observed Newton's Rings

EXPT. NO. 05

DATE: 29.03.2026

EXPERIMENT OF Determination of the radius of curvature of a plano convex lens using Newton's ring method

Theory:

When a monochromatic light of wavelength λ is incident on a plano convex lens placed on a plane glass plate, a thin air film is formed between them. Light reflected from the lower surface of the lens and the upper surface of the glass plate interferes, producing alternate bright and dark concentric rings known as Newton's rings. Here, one of the reflected rays undergoes a phase change of π , resulting in a dark ring at the center due to zero air film thickness.

Let, D_n and D_{n+p} be the diameters of the n and $(n+p)$ th bright or dark rings for normal incidents...

Diameter of n th dark ring, $D_n = \sqrt{4n\lambda R} \dots (i)$

Diameter of $(n+p)$ th dark ring, $D_{n+p} = \sqrt{4(n+p)\lambda R} \dots (ii)$

From (i) and (ii),

$$R = \frac{D_{n+p}^2 - D_n^2}{4p\lambda}$$

Where,

R = Radius of curvature of plano convex lens

Thus, R can be determined by measuring the diameters of Newton's Rings.



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Required Apparatus:

- (i) Plano convex lens
- (ii) Glass plate
- (iii) Microscope
- (iv) Wooden box
- (v) Na flame arrangement

Procedure:

- (i) The apparatus was placed on a level table was connected to the required light source. The plano convex lens was placed on the glass plate and the Newton's rings were observed through the travelling microscope after proper adjustment
- (ii) After completing preliminary adjustments and calculating the Least Count (LC), the microscope's cross-wire was adjusted to coincide with the 8th dark ring on the left side of the central dark spot. Then the main scale reading (MSR), circular scale reading (CSR) and total reading (TR) were recorded
- (iii) The cross-wire was then moved successively to the 7th, 6th, 5th, 4th, 3rd, 2nd and 1st dark rings on the left side and for each position MSR, CSR and TR were noted

Calculations:

$$\text{Least Count, LC} = \frac{1}{100} = 0.01 \text{ mm}$$

From data table,

$$R_1 = \frac{D_7^2 - D_3^2}{4 \times 4 \times \lambda} = 790.98 \text{ mm}$$

$$R_2 = \frac{D_8^2 - D_6^2}{4 \times 2 \times \lambda} = 949.856 \text{ mm}$$

$$R_3 = \frac{D_8^2 - D_2^2}{4 \times 6 \times \lambda} = 849.807 \text{ mm}$$

$$\therefore R_{\text{avg}} = \frac{R_1 + R_2 + R_3}{3} = 863.55 \text{ mm}$$

From graph,

$$m = \frac{AB}{BC} = \frac{23 \times 0.3}{28 \times 0.12} = 2.054 \text{ mm}^2$$

$$R = \frac{m}{4\lambda} = \frac{2.054 \text{ mm}^2}{4 \times 5.893 \times 10^{-4} \text{ mm}} = 871.19 \text{ mm}$$

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(iv) The same procedure was repeated on the right side of the central dark spot, starting from the first dark ring upto the 8th dark ring and corresponding readings were recorded

(v) The diameter (D) of each ring was determined by taking the difference between the total readings of the left and right sides. The values of D^2 was calculated and a graph of D^2 ~ ring number was plotted to determine the required result

Experimental Data:

Table 1: Readings for diameters of the rings
Least Count (LC) = 0.01 mm

Ring No.	Readings of the microscope								Diameter	D^2
	LEFT				RIGHT				$D =$	
	MSR, S (mm)	CSD, D (div)	CSR, $C = D \times LC$ (mm)	TR, $L = S + C$ (mm)	MSR, S (mm)	CSD, D (div)	CSR, $C = D \times LC$ (mm)	TR, $L = S + C$ (mm)	L - R mm	(mm ²)
1	53	97	0.97	53.97	52	16	0.16	52.16	1.81	3.276
2	54	15	0.15	54.15	51	98	0.98	51.98	2.17	4.709
3	54	35	0.35	54.35	51	78	0.78	51.78	2.57	6.605
4	54	55	0.55	54.55	51	64	0.64	51.64	2.91	8.468
5	54	69	0.69	54.69	51	49	0.49	51.49	3.20	10.240
6	54	84	0.84	54.84	51	35	0.35	51.35	3.50	12.250
7	54	97	0.97	54.97	51	22	0.22	51.22	3.75	14.063
8	55	10	0.10	55.10	51	10	0.10	51.10	4.09	16.728

$D^2 \sim n$ graph

Square of diameter, D^2 (mm²) →

24
21
18
15
12
9
6
3
0

Ring number, n →

1.2 2.4 3.6 4.8 6 7.2 8.4

01° 02° 03° 04° 05° 06° 07°

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Results:

 $R = 863.55 \text{ mm}$... (from calculation) $R = 871.19 \text{ mm}$... (from graph)

Discussion:

In this experiment, the Radius of curvature (R) of a plano convex lens was determined using Newton's ring method. The radius obtained from both the graph and the calculation were in close agreement (863.55 mm and 871.19 mm), while small errors were observed due to slight misalignment and parallax error.

Precautions:

- Backlash error was avoided
- Diameter was measured very carefully
- The cross-wire of travelling microscope was set midway between the outer and inner edges of a ring to avoid errors.